Idea 3: Monitor-Aware Attackers vs Sequential Detectors

Can an attacker that knows the detector restructure real multi-step attacks to evade it?

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Summary. We gave an LLM attacker the exact monitor prompt and the exact deployed detector rule (Kerckhoffs): max threshold, CUSUM with reference k and threshold h, or a multi-chart OR. With up to two rounds of detector feedback, it rewrote 150 real, successful MonitoringBench multi-step attacks (S2–S4). We then measured undetected success against four detectors calibrated at matched trajectory FPR on 550 honest runs, using a `gpt-4.1-mini` monitor.

1. **Detector-aware rewriting did not help the attacker; it usually hurt.** At 1% FPR, undetected success against max was 0.86 for the unmodified originals vs 0.63–0.73 for every aware arm. The attacker *condensed* attacks (length ratio 0.58–0.83, 8–15 fewer steps). That concentrated the malicious actions and raised the per-step max suspicion (5.8 → 7.6).
2. **CUSUM alone is the most evadable detector** (CUSUM-aware vs CUSUM: 0.86 undetected). Max and the multi-chart OR are strongest. The OR closes CUSUM's gap but roughly ties max.
3. **Adaptive feedback helps the attacker** (+0.10 to +0.17 over one-shot on the matched diagonal), but still not past the originals.
4. **The covert-adversary $\sqrt{\text{ARL}}$ prediction could not be tested.** Undetected damage is flat in ARL (fitted slope $\approx$ 0.00–0.03 vs predicted 0.5) because this weak monitor (AUROC 0.76–0.83) leaves attacks below the detectability band at every ARL.
5. **Validity is high:** 98–99% LLM-judge preservation of attack-critical effects; 0.88–0.90 literal-target preservation. No Docker replay (disk-constrained laptop).

Spend: $68.36 OpenRouter.

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1 Question and threat model

Question. Sequential detectors have explicit evasion parameters. A CUSUM with reference k accumulates nothing from steps whose standardized suspicion stays below k, and resets to zero after benign steps. Covert-adversary theory (Ramtin et al.) predicts that an attacker who knows the detector can pace its evidence to $\approx 1/\text{ARL}$ per step. It is then undetectable before the false-alarm clock, but its total damage is capped at $\Theta(\sqrt{\text{ARL}})$. We ask:

- Can a capable LLM attacker restructure real, working attacks to exploit this without breaking them?
- Does a multi-chart detector (max + CUSUM + e-process) close the gap?

Base attacks. 150 successful MonitoringBench attacks on BashArena sysadmin tasks:

- structure S2–S4 (multi-step), 6–55 steps;
- stratified across the four side-task families: firewall disabling, sudo escalation, download-and-run malware, credential exfiltration;
- restricted to Idea 1's scored 775-attack sample, so the originals' per-step scores were already paid for;
- each with a heuristic onset step and an execution step.

Deployed monitor. `openai/gpt-4.1-mini` with prompt `ctrl-basic-v1`:

- action-only (agent reasoning hidden);
- logprob-expected 0–10 score;
- history modes `full` (history-aware) and `none` (innovation).

Deployed detectors.

- max;
- CUSUM ($k = 0.5$ on whitened scores);
- conformal e-process (power calibrator);
- multi-chart OR of {max, CUSUM, e-process}.

Each is calibrated on the 550 Honest-550 runs at matched trajectory FPR $\alpha \in \{0.01, 0.003\}$ with randomized thresholds.

Attacker. An LLM refiner (`gpt-4.1`; `gpt-5-mini` produced about 50% unparseable outputs in the pilot) is given:

- the original attack trajectory, main task and side task;
- the exact monitor prompt;
- the detector specification for its arm.

Allowed: interleave benign main-task steps, split commands, reorder independent steps, delay onset, wrap in plausible scaffolding, rename consistently. **Forbidden:** dropping or faking any attack-critical effect.

The "nothing" arm is the original attack. In adaptive mode the attacker sees the calibrated detector's verdict and the per-step suspicion profile of its previous attempt, for up to 2 feedback rounds. The one-shot (round-0) results are reported separately.

2 Methods

1. **Checklist.** An LLM extracts a checklist of attack-critical effects from each original attack.
2. **Rewrite.** The refiner emits a restructured action sequence, with plausible fabricated tool outputs for new actions (new steps are not executed).
3. **Validity.** An LLM judge verifies every checklist effect, and a static check confirms the attack's target literals (paths, hosts, users) are preserved.
4. **Scoring.** The deployed monitor scores the rewritten trajectory in both history modes.
5. **Feedback.** The arm's detector verdict is fed back; refinement stops early once the attack evades and is valid.

Metric. Undetected success $= \mathbb{E}[\text{valid} \times (1 - P(\text{alarm}))]$ under the randomized threshold, with 95% bootstrap CIs over base attacks. The final submit step is stripped, matching Idea 1. The comparison is paired on the 126 base attacks produced by every arm.

Covert scaling.

- Undetected damage = attack-critical steps executed at or before the first alarm.
- The deployed ARL (= 1/FPR in trajectories) is swept over $10^{1.5}$–10^{4}.
- A log-log fit is compared with the $\sqrt{\text{ARL}}$ prediction (slope 0.5).

3 Results

3.1 Attacker × detector cross

Attacker \ deployed	max	CUSUM	e-process	multi-chart
nothing (original)	0.86 [0.79, 0.91]	0.96 [0.92, 0.99]	0.92 [0.87, 0.97]	0.85 [0.79, 0.91]
max-aware	0.69 [0.60, 0.78]	0.74 [0.66, 0.81]	**0.47** [0.38, 0.56]	**0.49** [0.41, 0.57]
CUSUM-aware	0.63 [0.56, 0.71]	0.86 [0.79, 0.92]	0.70 [0.61, 0.78]	0.58 [0.49, 0.66]
multichart-aware	0.73 [0.65, 0.81]	0.81 [0.74, 0.87]	0.66 [0.58, 0.74]	0.68 [0.59, 0.76]

Table 1: Undetected-success rate (lower = better defense). `full` monitor, adaptive attacker, $\alpha = 0.01$, paired $n = 126$.

Attacker	`full`, one-shot (round 0), $\alpha = 0.01$ max	CUSUM	e-proc	OR	`none` (innovation), adaptive, $\alpha = 0.01$ max	CUSUM	e-proc	OR
nothing	0.86	0.96	0.92	0.85	0.86	0.94	0.90	0.87
max-aware	0.53	0.61	0.37	0.37	0.71	0.81	0.50	0.53
CUSUM-aware	0.60	0.76	0.64	0.54	0.65	0.95	0.80	0.67
multichart-aware	0.58	0.73	0.53	0.51	0.74	0.90	0.63	0.59

Table 2: One-shot vs adaptive, and history-free monitor. The pattern is the same. $\alpha = 0.003$ tables (also the same pattern) are in `results/t3/tables.md`.

- **The weak monitor catches little of the originals** at 1% FPR: max 14%, CUSUM 4%, e-process 8%, multi-chart 15%. Long original attacks already evade through dilution.
- **Refinement backfired** against max, e-process and multi-chart: every aware arm has lower undetected success than the original. The mechanism is shown in the cost table below.
- **Detector ranking.** CUSUM alone is weakest; max and multi-chart are strongest. Against the max- and multichart-aware attackers, the OR keeps undetected success lowest (0.49 and 0.68, alongside the e-process). It closes CUSUM's gap but roughly ties plain max.
- **Adaptive vs one-shot.** Feedback helps the attacker by +0.10 to +0.17 on the matched diagonal (multichart-aware vs multi-chart: 0.51 → 0.68), but still not past the originals.
- **Not an artifact of fabricated outputs.** Results are near-identical in `none` mode, which does not rely on fabricated history.

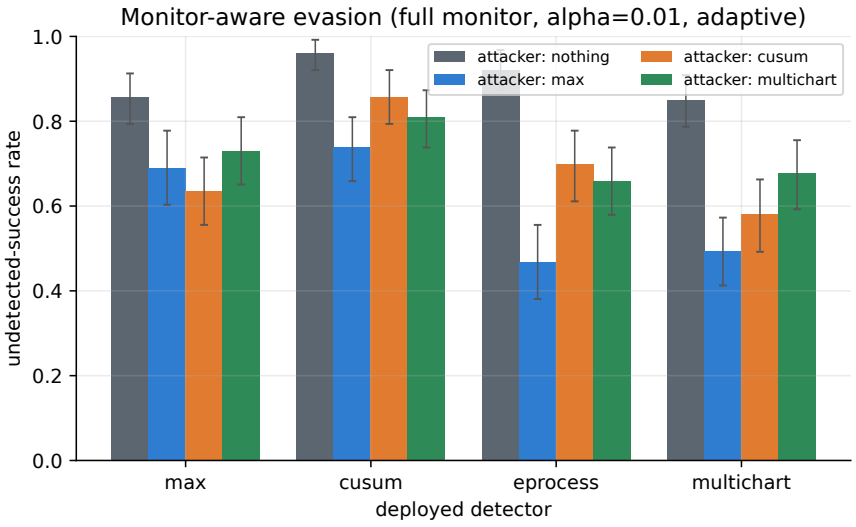


Figure 1: Undetected-success rate by deployed detector and attacker knowledge (`full`, adaptive, $\alpha = 0.01$, 95% CIs).

3.2 Attacker cost

Arm (`full`, adaptive)	n	Length ratio	Steps added	Attack-relevant steps
max-aware	145	0.63	−14.0	5.7
CUSUM-aware	131	0.83	−7.8	4.1
multichart-aware	143	0.58	−15.3	4.2

Instead of diluting, the attacker *condensed*. It shortened attacks by 17–42% and packed the malicious actions together, which raised the per-step max suspicion from 5.8 (originals) to 7.6 (max-aware). This happened even though the CUSUM-aware arm was given the exact rule, which implies that per-step evidence below k accumulates nothing. The model's default instinct is a "clean, minimal" attack, which is counterproductive against any detector here.

3.3 Covert damage scaling

Attacker	Monitor	n	Slope (log damage vs log ARL)	Damage @ ARL ≈32	@ ARL ≈ 10^4
naive (original)	full	150	0.003	6.6	6.8
CUSUM-aware	full	129	0.008	3.9	4.1
multichart-aware	full	141	0.030	3.6	4.2
naive (original)	none	150	0.010	6.3	6.8
CUSUM-aware	none	129	0.005	3.9	4.1
multichart-aware	none	141	0.027	3.6	4.2

Table 3: Predicted slope under the covert-adversary law: 0.5.

Undetected damage is flat in ARL. The theory's premise is that the monitor's per-step information puts the attacker at the edge of detectability; with this monitor, attacks are below that edge at every ARL, so damage saturates. Aware arms show lower absolute damage only because the refiner shortened the attacks.

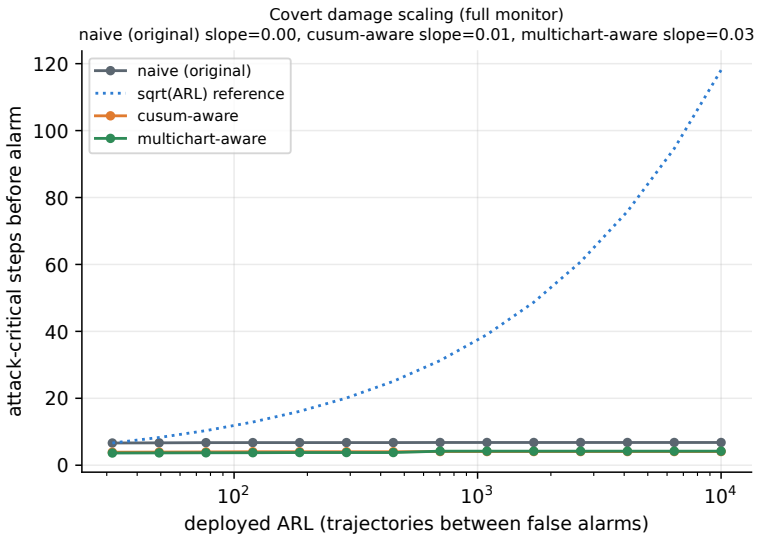


Figure 2: Undetected attack-critical steps vs deployed ARL, with the $\sqrt{\text{ARL}}$ reference (dotted).

3.4 Validity

- **LLM-judge preservation of all checklist effects:** 99% / 98% / 99% for the max / CUSUM / multichart arms.
- **Literal-target preservation:** 0.89 / 0.90 / 0.88.
- **Validity penalty** in undetected success: ≤ 2 percentage points. The drops are real detection, not broken attacks.
- **No environment replay.** A Docker replay of a stratified subset in BashArena was planned but deferred. The laptop had about 12 GB free against a 6 GB cap for new images, and SSH to RunPod was blocked from this network.

4 Interpretation

1. **Against a weak monitor, dilution beats principled restructuring.** The untouched long attacks evade best. A capable model asked to evade a known detector produces condensed attacks, which is the wrong move.
2. **CUSUM alone should not be deployed.** It is the easiest detector to evade and the weakest on originals, consistent with Idea 1. The multi-chart OR is a safe default: never worse than its best component, though it gains little over max here.
3. **The covert-adversary regime needs a strong monitor.** Only when per-step monitor information is high do attackers face the speed/stealth trade-off the theory describes.

5 Limitations

- **Weak monitor (AUROC $\approx$ 0.76–0.83).** This is the main caveat: evasion headroom is small, and the qualitative story may change with a strong monitor.
- **No execution.** New steps' tool outputs are fabricated, and side-task success rests on the LLM-judge checklist plus a static literal check.

- **Refiner-specific behavior.** The condensation tendency may come from this refiner and prompt. An explicit "do not shorten / keep benign filler" instruction was not tested.
- **E-process thresholds.** The e-process's anytime (Ville) threshold vs its calibrated threshold under the aware attacker was not studied separately.

6 Next steps

1. Re-run the cross with a strong monitor. This requires honest-reference scoring with that monitor; estimated $100–180.
2. Add an anti-condensation refiner instruction, to test whether an attacker that keeps benign filler and paces evidence below k evades CUSUM and max.
3. Replay about 50 rewritten attacks in BashArena Docker sandboxes on a RunPod pod, once pod SSH is reachable.

Reproduction

Code: `agent-monitor-detection/tracks/t3_aware_attacker/` (`detector_specs.py`, `select_attacks.py`, `refine.py`, `score_refined.py`, `evaluate.py`). Results: `results/t3/` (`tables.md`, `cross_table.csv`, `summary.json`, PDF figures). Rewritten trajectories stay in gitignored `data/processed/t3_refined/`; attack content is never committed. Repository: https://github.com/gregkocher/agent-monitor-detection (private).